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The evolution of patent system: invention created by artificial intelligence

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Abstract

With development and application of computer science research in depth, the invention created by Artificial Intelligence (AI) could fully achieve the standards of granting patents. The paradigm shift brought about by the scientific revolutions cannot be separated from the constraints of social systematic structure. Even if these inventions created by AI will not have an excessive impact to against our patent system, the general public's concerns about technology out of control have led to the continuous delay of patentability of the inventions created by AI. From the principle of interest balance to deal with these problems, it is only necessary to coordinate and resolve the legal relationship behind the inventions created by AI, and make appropriate adjustments through patent system. To give AI inventors the personality status through legal fiction will not only not destroy the ethics and rules of human society, but will be more conducive to stimulating the research and development of AI research and further promoting its industrialization.

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1. Introduction

Artificial intelligence (AI), also known as machine intelligence, refers to a machine made by human beings that shows a method of intelligent autonomous production. Its core is to simulate and expand people's thinking abilities

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such as reasoning, knowledge, learning, perception, planning, communication and operation, and even beyond the limit of the human brain can solve problems. With the breakthrough of core algorithms and the basic support of big data, the field of AI has ushered in a period of sustained and rapid growth. In particular, the in-depth integration of AI technology and life sciences continuously has led to a significant reduction in the cycle of technological iteration. The way in which human beings participate in innovative activities is changing, and AI has become a new source of creativity. In essence, the new scientific revolution is a worldview transformation related to social consensus, which will inevitably bring many challenges to the current legal system and scientific ethics, and even bring about a subversive revolution. Among them, the first legal system revolution will occur in the patent law, which is most closely related to innovation activities.

Since the concept of "artificial intelligence" in the modern sense was first proposed in 1956¹, the research ideas of AI have been attached to computer science, that is to realize the autonomous operation of machines through algorithm construction. With the increasing demand from the market and society for AI to handle more complex tasks, the research on AI has begun to focus on how human beings generate intelligence, and then make machines learn how to think and behave. At this time, the field of AI is no longer a single computer science, but integrating biology, medicine, cognitive neuroscience, psychology and other disciplines to carry out interdisciplinary research. The technological development of AI seems to be moving from an instrumental human-like machine to a superhuman intelligence with autonomous thinking, which will break the foundation of the human-dominated social structure. In the process of R&D, if we do not consider the impact of AI on legal system and technology ethics, it will inevitably lead to *The Problem of Frankenstein*².

Intellectual property (IP) system is the basic law of technological innovation. The conflicts of AI technology on the structure of human society are first reflected in the IP laws, and its history can be traced back to the early development of computer science. In 1965, the first work considered to be independently created by a computer was submitted to the Register of Copyrights in the United States³. How to deal with the dispute over the uncertain legal subject status of AI has led to different forms of challenges in conceptual interpretation of legal terms. And these problems that subvert cognition have continued to this day. Along with the technologies of artificial neural network, machine learning and human-machine interaction have achieved breakthrough progress, if it is said that the challenge to copyright law only stays at the level of expression innovation, the challenge to the patent law will further reflect that the innovation of "thinking" is no longer a patent of mankind. For inventions assisted by AI, a large amount of human intervention and guidance is required. There is no dispute regarding the application of the current patent system on the issue of obtaining patent protection. The invention created by AI is produced autonomously without human intervention. In other words, when human beings are liberated from R&D activities, should the patent law recognize the alternative innovation of AI, and should the resulting rights and interests be allocated to the "inert" inventor who create or own the AI?

In fact, the inventors who created the Above-Human AI are often in organized entities such as large technology companies and R&D institutions. Compared with these monopolistic scientific research organizations, it is obviously unfair if the patent law treats them equally with ordinary technology research and development personnel. Furthermore, if an invention created by AI is granted a patent right, how to determine the scope of legal protection and examination standards? Based on the above-mentioned institutional impact and theoretical challenges brought about by the development of AI technology, it can be further summarized into three open questions:

- The legal subject status of the invention created by AI, that is, how to treat the relationship between general inventor, inventor of AI and AI as inventor, and whether the patent system that pursues the principle of "equity of inventors" should be protected separately;
- The scope of patent rights for the invention created by AI, that is, how to determine the examination standard, scope of authorization, and the scope of restrictions;

¹ In fact, different ancient civilizations have recorded similar concepts as "artificial intelligence", and the core idea is to replace human manual labor with machines.

² As the British writer Mary Shelley described in her book, a scientist named Frankenstein created a new humanoid life form, which triggered a series of criminal and anti-human events. This novel, completed in 1818, reveals the uncontrollable consequences of uncontrolled technology and calls on people to pay more attention to the nature and society in which we live when engaging in scientific research and development.

³ See Register of Copyrights, Sixty-Eighth Annual Rep. of the Register of Copyrights 5 (1966).

- The ownership distribution of the invention created by AI, that is, how to distribute and redistribute benefits among inventors of AI, owners of AI, and the public.

We will analyze these issues, and then respond to the impact of AI against law and ethics. It is true that the openness of these issues will inevitably lead to the diversification of understanding and response. This article aims to seek a delicate system balance between the social justice and social efficiency from innovative activities of AI.

2. Subject status of AI inventor in patent law

In the past practice of patent applications, it seems that only the object standard of the invention is used as the sole criterion for granting patents, while the subjectivity status of the inventor is often ignored. This means that the subject engaged in innovative activities is not limited by age, intelligence and other mental conditions. If the subjectivity of the inventor is not considered, the invention created by AI can undoubtedly be granted patent rights. However, it is easy to cause the subject confusion between human and AI, which will hinder the institutional function of the patent law that should encourage most people to engage in innovation. In other words, when it is not disclosed that an invention created by AI, it may actually be patented by those who get it for nothing because the invention achieves appearance characteristics of patent. On the other hand, if we adopt a conservative approach that excludes the completion of inventions created by AI from patent law, it seems that the above phenomenon cannot be avoided. At the dawn of the intelligent society, the patent system should take seriously the market benefits and technological value which created by AI. It is necessary to adopt an active and diversion strategy to solve the problem of interest distribution and legislative systemization, rather than avoiding and blocking.

2.1. Justifying Entitlement for AI

At present, countries like Saudi Arabia that Sophia, a humanoid robot, who have been granted citizenship directly are only cases. In the entire legal system, there are still many obstacles to recognize that AI enjoys the same rights, duties and responsibilities as human beings. However, prioritizing individual legislation (especially IP system) to break through the issue of the qualification of AI as a legal subject can be regarded as a paradigm shift revolution with more positive externalities. In recent years, the United Kingdom Patent Office (UKPO), the European Patent Office (EPO) and the United States Patent and Trademark Office (USPTO) have rejected patent applications successively that designate AI as the inventor. Under the current legal system, only human beings can apply for and hold patents. The patentability problem of inventions created by AI seems unsolvable.

From the perspective of institutional economics, the essence of IP system is to construct the business model which is most conducive to innovation activities, thereby reducing the transaction cost of technological innovation. As invention created by AI may become more and more common, the patent system must be at the forefront of the development of AI technology, in order to stimulate and accelerate the process of transformation to an intelligent society. In fact, according to the perspective of utilitarianism, granting patents to the invention created by AI is not to subvert the human-centered social order, but to achieve a fairer distribution of benefits. Assuming that the invention created by AI is not protected through patent law, it is extremely likely that the forgery of a human inventor who will improperly monopolize the patents under the drive of maximizing market profit. Therefore, in order to ensure the healthy and sustainable development of AI industry, it is an economic rationality to recognize the patentability of inventions created by AI from the institutional level.

From the deontological views, it seems difficult to accept that AI as a legal subject equivalent to human beings. This is tantamount to a human self-revolution. This is tantamount to a human self-revolution. However, the intelligence of AI is not created out of thin air, including algorithm engineers who set the policies that dictate how the machine thinks, data programmers who set the parameters of the database used to train the AI machine and investors. Their contributions have endowed the creativity to AI. Therefore, the patent system should not ignore the contribution of such a large innovation team to social innovation. Furthermore, granting patent rights to the

invention created by AI is essentially a way similar to "shooting a movie", which enables different contributors to obtain the right to secondary remuneration⁴.

2.2. Method by Legal Fiction

In the provisions of the legal subjects, the inventor and the patentee can be separated. According to Implementing Regulations of the Patent Law of PRC, "inventor" or "creator" referred to in the Patent Law means any person who makes creative contributions to the substantive characteristics of an invention-creation. It is generally believed that the "person" here can only be a natural person. Even for service inventions, the patentee can be a legal person or other organization, but the inventor can only be a natural person. At the same time, the Chinese Patent Law requires that the name of the inventor should be stated in the patent application document, and the inventor has the right to state that he is the inventor in the patent document. In addition, the legal provisions of different jurisdictions including the European Patent Convention and the United States Patent Law, as well as judicial precedents⁵, all express or imply the idea that people are the subject of invention-creation. However, as far as the invention created by AI is concerned, human creativity is not directly attached to these inventions, and human beings cannot accurately predict what specific kind of invention AI can create.

An invention-creation is the result of intellectual labor and are not restricted by civil capacity. In the activities of invention-creation, no matter whether the person who completes the invention creation has complete civil capacity, he can be recognized as the inventor as long as he has completed it. For example, in the Chinese Patent Law, "invention" refers to new technical solutions proposed for a product, a process, or the improvement, and do not impose any restrictions on the subject of the invention-creation. This means that the invention created by AI cannot be excluded from patent protection as long as they comply with legal requirements. In addition, after the invention is completed, if the patent application right is enjoyed by others, then the inventor does not need to perform other obligations according to the provisions of the Chinese Patent Law (but according to the U.S. Patent Law, the inventor needs to swear or declare). Therefore, there are no obstacles to identifying AI as an inventor at the legal and practical levels. As for the secondary remuneration rights enjoyed by programmers and investors who create AI, it will not hinder granting AI the status of inventor.

From the perspective of legal theory, as long as the legal procedures are passed, the law can draft new legal subjects and give them corresponding rights, such as corporate juridical person. Even if the inventor is interpreted in a narrow sense, there is the possibility of fictionalizing the legal personality or subject status in the legal sense for AI. Legal fiction is irreplaceable because it does not violate the cognitive logic of reality and the existing legal system, which makes the internal distribution of patent rights effectively.

2.3. Distinguishability of AI Inventor

Comparing patent and copyright, there are differences between inventions and works which both created by AI. A work is a personalized expression of aesthetics, imagination, and expressiveness, so there are more personal and moral rights led to the principle of automatic protection without any formalities in copyright law; while an invention a technical tool, plan, and method discovered in the process of mankind's transformation of nature. So, the conditions for granting a patent require that the invention does not belong to the prior art and must go through the authorization procedures for application approval. In other words, unlike authors who enjoy permanent personality rights, actual inventors only show the factual identity of creation, and do not necessarily lead to the formation of legal rights, duties and responsibilities. At the same time, the inventor has the personal right to sign on the patent document to demonstrate the personality relationship between the inventor and the invention. This personal right is permanent, renounceable, and non-transferable, and has no substantive significance in determining the property

⁴ The copyright of film works is exercised uniformly by the producers, but the copyright of music, art, photography, script and other works in the film can be exercised independently. The right to secondary remuneration of film refers to the main creators of the film can obtain additional rewards in addition to the basic remuneration and their own intellectual property rights.

⁵ See *University of Utah v. Max-Planck-Gesellschaft*, Fed. Cir. Nos. 2012-1540, -1541, -1661 (August 19, 2013).

ownership of the patent right. Therefore, the patent law's principle of distinguishing inventors, applicants and patentees makes AI fictionalized as legal personality not impact the existing patent law.

On the other hand, AI, as the "object of the machine and tool used in the invention" and "the fictitious subject of the invention", has a legal status distinction, which affects the design of specific rules of the patent system. Once AI is drafted as an inventor, the inventor must be divided into "natural person inventor" and AI inventor. At this time, inventors in the sense of the patent law are no longer limited to natural persons, and "persons who have made creative contributions to the substantive features of the invention" also include AI. As an inventor, AI can be modeled as a single inventor who generates an invention independently, or it can "co-create" an invention with a natural person through the identity of a "co-inventor". Technical means should be adopted to mark inventions created by AI to prevent counterfeiting of human invention patents. For invention patents created by human-machine cooperation, participants should be required to submit an explanation of the inventor's actual contribution and a statement of willingness to accept stricter rights restrictions.

3. Patentable scope of invention created by AI

In order for an invention to be protected by the patent law, it must meet the object examination standard set by patent law. First of all, the invention must fall within the scope of patentability subject, that is, the invention does not belong to the technical exclusion field stipulated in patent law. Then, the invention must have the object standard of patentability, that is, novelty, creativity and utility. Based on the system pursuit of the value goal of technology being good, the invention created by AI should also be restricted by scientific ethics.

3.1. Standards of Patentability

The technological achievements created by AI must follow the general provisions of the patent law, including the object type, application procedure, examination standard, scope of authority, etc. For novelty examination, AI can search and crawl massive existing technologies in patent database, and then avoid the risk of violating novelty and even infringement. However, the use of big data technology by AI will also go to another extreme, that is, AI can use the documents, papers and technical information that have been published on the network for technical improvement. These public technical solutions may not meet the patent application standards or are in the initial stage of research and development. How to avoid this unfair innovation competition will become a key issue in the future patent system.

Judging from the standard of creativity examination, AI could create inventions with substantive characteristics and significant progress, but it cannot rule out the generation of a large number of low-quality patents. Once the patentability of the invention created by AI is recognized, it is bound to put forward higher requirements for the system carrying capacity and examination efficiency of patent examination institutions. On October 6, 2020, USPTO released the Public Views on Artificial Intelligence and Intellectual Property Policy⁶. The report points out that the vast majority of public opinions indicate that the invention created by AI will affect the judgment standard of the patent office and the court for the professional level of "technical personnel in the field". This relatively subjective examination standard will directly affect the determination of patentability. Therefore, even if the market and industry have great interest demands for the "creation" of AI inventions, are patent administration departments and relevant examiners really ready for the arrival of the era of AI?

Generally speaking, patentability is not affected by the process, mode and way of invention. As long as an invention has prominent substantive characteristics and significant technical progress compared with the existing technology, it can apply for a patent. However, as far as utility examination is concerned, there may be licensing barriers for inventions "created" by AI in some countries. Taking China's patent examination standard as an example, it is required that the technical scheme should not violate the natural law and should have reproducibility. The so-called "reproducibility" refers to that according to the technical scheme steps specified in the open patent application documents, technicians can repeatedly implement and manufacture the same technical content. But, the creation's

⁶ See United States Patent and Trademark Office, Public Views on Artificial Intelligence and Intellectual Property Policy, available at: https://www.uspto.gov/sites/default/files/documents/USPTO_AI-Report_2020-10-07.pdf, last visited Oct. 29, 2020.

process of AI is based on the abstract operation of algorithmic program and code language. Even the engineers who create AI cannot copy the invention steps derived from algorithm calculation. Therefore, under the utility characteristic of patent, human intervention and participation are needed in the process of AI invention, so as to ensure the descriptive writing of patent application documents and the reproducibility of technical solutions, and at the same time, it can avoid the danger of technology out of control to human society.

3.2. Exclusion of Technology

The types of inventions that are not patentable stipulated in the patent law are essentially the scope of patentability from the opposite side. In the practice of patent examination, priority is usually given to judge whether an invention belongs to the legal type of technology excluded by the patent law. The reason why the patent system excludes these types of technology is not that they do not have the three characteristics that constitute patents, but there are different patent policy objectives. The invention created by AI naturally follows the limitations of these technical exclusions.

However, at the initial stage of the development of AI technology, the general public has not reached a social consensus on the patentability of the invention created by AI. And there are still postmodern anxieties such as the black box of AI technology and algorithmic bias. In the procedural stages of patent examination, objection and invalidation, there may be the extension or analogy of the invention created by AI as "rules and methods of intellectual activity" or "scientific discovery". It is true that such an explanation goes against the cognitive logic of reality, and also greatly hinders the entry of more abundant technological products into the market and public domain. It is better to reform a few specific rules of the patent system in exchange for an open and healthy technological innovation environment, rather than acquiesce in a conservative mindset that inventions created by AI flow into the "underground market" or counterfeit inventions created by human beings.

3.3. Limitation of Technology Ethics

The dilemma of patent law for the invention created by AI is essentially the concern of human beings about the impact of AI on the development value and rules of human society. In the patent system, this worry is that AI will replace human's innovation ability, which is an important feature that distinguishes human beings from other life bodies. In order to balance the development consistency of AI technology and social cognition, including AI research and development personnel, industry associations, intergovernmental organizations and other stakeholders put forward the norms and explanations of AI application ethics at different levels. Among them, the most representative one is the Preliminary Draft Reports on COMEST on Robotics Ethics jointly issued by the United Nations Educational, Scientific and Cultural Organization (UNESCO) and World Commission on the Ethics of Scientific Knowledge and Technology (COMEST) in 2016⁷, which determines that in addition to following the basic ethics of human society, AI also needs to write specific ethical guidelines into the development of AI. According to the European Commission's Report with recommendations to the Commission on Civil Law Rules on Robotics⁸, it is pointed out that the automation trend brought about by AI may pose challenges to safety and ethics, as well as a series of legal issues, such as personal information and privacy protection, adjustment of employment relationship in labor market, robot infringement and liability.

In the process of application, implementation and judicial relief of the invention created by AI, the restriction rules of scientific ethics play an irreplaceable role. As far as the patent system belongs to the human society, even if the AI inventor is supposed to be a legal personality, the benefits generated by his patent right still belong to human beings. Therefore, the anthropomorphic right enjoyed by AI is actually the right of human beings, and the interest groups behind AI should also be the subject of legal responsibility. This reflects that the development of AI

⁷ See United Nations Educational, Scientific and Cultural Organization & World Commission on the Ethics of Scientific Knowledge and Technology, Preliminary Draft Reports on COMEST on Robotics Ethics, available at: <http://unesdoc.unesco.org/images/0024/002455/245532E.pdf>, last visited Oct. 29, 2020.

⁸ See European Commission, Report with recommendations to the Commission on Civil Law Rules on Robotics(2015/2103(INL)), available at: <http://www.europarl.europa.eu/sides/getDoc.do?type=REPORT&mode=XML&reference=A8-2017-0005&language=EN>, last visited Oct. 29, 2020.

technology cannot be separated from the influence of human beings. It is guided by human influence and guided and controlled by the ethical rules of human society. In the process of patent system reform, we should pay more attention to the principle of good faith and public order and good custom reflected in the law of science and technology ethics, that is, the applicant should honestly disclose the type, process and content of the invention created by AI, and cannot apply it to the field with safety risk.

4. Patent rights allocation for invention created by AI

The key of AI to maintain "intelligence" is the continuous improvement of algorithm, which requires human to continuously invest huge infrastructure elements such as intelligence, capital, energy and so on. After the invention created by AI is patented, it is necessary to analyze the different stakeholders and their interest demands. Based on the "participation contribution" such as intelligence elements or capital elements, the researchers, owners and users of AI have legitimate interest claims for the intangible property generated by the results of AI. When its AI generated production involves infringement disputes, it is necessary to have a qualified legal subject to protect the rights and interests and bear the responsibility.

4.1. Algorithm Engineer of Building AI

The core algorithm engineer of AI determines the way and strategy of machine thinking, and is the person who contributes the most creative intelligence elements. Although the invention created by AI cannot directly show the wisdom of Algorithm Engineers, it runs according to the program they create. In other words, algorithm engineers provide the "brain" for AI. However, for the patents of AI, the wisdom contributed by Algorithm Engineers only stays at the "thinking" level, and no specific and definite technical scheme has been formed. For the intellectual labor paid by Algorithm Engineers, the first profit distribution has been carried out in the form of copyright and labor remuneration for the algorithm program of AI. As human subjects who have contributed enough intelligence elements, the value of intellectual labor of algorithmic engineers should be given more property rights incentives. Therefore, in the process of interest distribution of inventions created by AI, they should be given the claim to secondary remuneration.

4.2. Database Programmer for Machine Learning

The database maintenance programmer of AI sets the data parameters of machine learning, which ensures the ability of AI to obtain and transform information content. Big data provides innovative nutrients for AI, and data programmers are equivalent to Librarians in "big data library", maintaining the external environment for autonomous learning of AI. Similar to the legitimate reason for Algorithm Engineers to obtain the right to secondary remuneration, data programmers pay a lot of intellectual labor in the process of managing and maintaining the basic database, and they are also the supervisors and controllers of AI in the process of creative activities.

4.3. Investor/ Owner of AI

The owner of AI, that is, the investor of AI, is the legal person or other organization that invests in the development of AI. In the process of developing AI technology, the owners organize and invest more costs and resources, and actually bear the legal risks of its patents. Through the investment in AI research and development, the owner who takes commercial profit as the primary goal finally realizes the social benefits of promoting technological innovation and progress. The modern patent system allows the separation between the inventor and the patentee, the most typical of which is the service invention system. Taking the provisions of the Chinese patent law for service invention as an example, the right to apply for a patent, the right to apply for a patent and the right to patent belong to the unit where the service inventor belongs. For the invention created by AI, the owner of AI has the right to apply for patent according to the provisions of service invention, and then perform the right of patent application, and enjoy the patent right after the patent is authorized. The research and development of AI is a form of scale economy, which can not be separated from a large amount of funds and organized investment of

intelligence. From the perspective of institutional economics, the recognition of investors as the main body of patent rights can reduce the market transaction costs and realize the optimal allocation of technological innovation efficiency.

4.4. User and General Public

The patent system is a tool to achieve the purpose of public policy, so we need to pay attention to and adjust the social externality brought by property right. Based on the consideration of protecting the public interest, setting strict restriction system on the invention created by AI not only has moral legitimacy, but also limits the blind expansion and improper use of patent right from the purpose of stimulating innovation. In the invention process of AI, it is easy to occupy too many public information resources, and even take others' immature technical solutions as their own. In order to avoid the haze, this kind of free and unrestricted information abuse behavior, we should not give too much exclusive rights to the invention created by AI, but should pay more attention to the sharing of technological innovation.

If the property right of invention created by AI is for the purpose of stimulating researchers and investors to make initial distribution, then the public domain reservation is an important reason to realize the redistribution of interests by improving the patent restriction rules. To recognize the patentability of the invention created by AI, it is necessary to clarify the specific time limit, scope of patent rights, licensing methods, compulsory licensing and other supporting rules, so as to promote the endogeneity of technological re innovation.

5. Conclusion

The invention created by AI is essentially an information product, which has the legitimacy of property in the market transaction. We should rely on the patent system to coordinate the value balance between trade freedom and transaction security proposed by the innovation market. The invention created by AI is the product of systematic innovation, involving multiple stakeholders. At the same time, we should take the different demands of multiple stakeholders seriously, and construct the interest distribution mechanism of the invention created by AI under the normative framework of patent system, which will not destroy the legal system and moral ethics that the social order depends on, but also help to stimulate the endogenous innovation of AI research and industry.

The patent system should take a correct normative approach to the invention created by AI: on the one hand, it is necessary to clarify the patentability of its invention as soon as possible, so that the interests of all parties can be properly returned, distributed and coordinated under the property right system; on the other hand, we should see the special problems of the invention created by AI in the process of legislative, administrative and judicial. In order to reduce the impact of technology out of control on the ethical basis of human society, it is necessary to strengthen the examination standard of scientific ethics. In a word, the patent system should have a sense of innovation, extend the principle of balance of interests between human beings, nature and society to the field of intelligent machines, and realize the sustainable development of human-computer collaborative innovation in the coming intelligent era.

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